Automating Mammogram Reports Using Large Language Models for Standardized, Clinically Actionable Insights

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Contents

- Breast cancer is one of the leading causes of cancer-related deaths worldwide.
- Mammograms are critical imaging studies for early breast cancer detection.
- Clinical reports contain important diagnostic descriptors such as mass shape, density, and microcalcifications.
- Most mammogram reports are written in unstructured free text format.
- Unstructured reports make it challenging to perform large scale data analysis and automated AI tasks.

Why Structured Reports?

- **Data Analytics:** Structured reports enable easier aggregation and analysis of clinical findings.
- **Decision Support:** Helps clinicians make faster and more consistent diagnostic decisions.
- **AI Applications:** Supports downstream tasks like predictive modeling, classification and research.
- **LLM Potential:** Large Language Models can standardize unstructured reports into structured clinical data.

Research Gap & Aim of the Paper

- Most prior work focuses on imaging feature extraction or lesion detection, not natural language report generation.
- Existing feature extraction methods do not leverage the generative and contextual reasoning capabilities of modern LLMs.
- Also current methods are largely manual and unstructured, preventing the realization of potential time and cost savings.
- **Aim:** Develop an LLM based pipeline to generate mammogram reports grounded in structured features and evaluate them quantitatively using a carefully annotated CSV file.

Theoretical Background

We found that these mammographic features are used to effectively detect breast cancer from mammograms:

- Mass shape and microcalcification patterns are correlated with malignancy risk.
- Architectural distortion is a key mammographic sign associated with malignancy.
- Breast density influences mammogram sensitivity and cancer risk stratification.

LLMs provide contextual understanding to generate coherent narrative reports from extracted features.

Theoretical Background

The following features were used for report generation and evaluation:

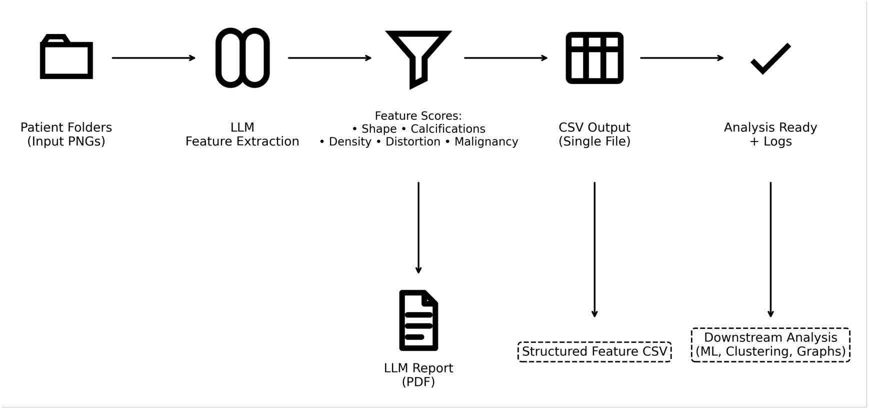
- mass shape
- microcalcification pattern
- architectural distortion
- breast density category
- malignant

Methodology

We propose an LLM based pipeline for structured mammogram report generation from imaging data. The workflow consists of the following steps:

- **Data Input:** Patient folders containing mammogram images are provided as input to the system.
- **LLM Guided Feature Extraction:** The LLM's vision-language module extracts relevant mammographic features from input images and converts them into structured data.
- **Report Generation:** The LLM applies contextual reasoning to interpret the extracted features and generate structured, clinically coherent mammogram reports in PDF format.
- **CSV Output:** The extracted features are also exported as a CSV file for record keeping and further analysis.

Methodology

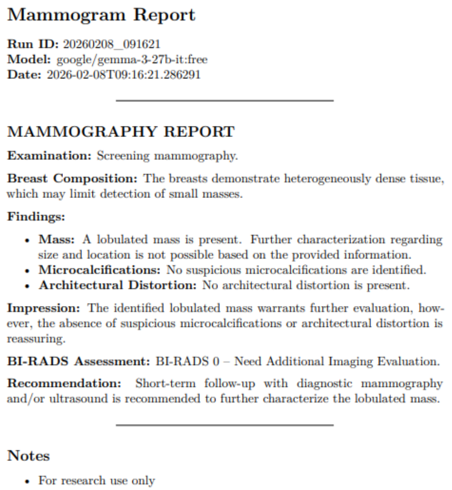


End-to-end pipeline showing guided feature extraction, LLM based report generation and CSV based evaluation.

Methodology

```
mammogram_function = {
    "name": "generate_mammogram_report",
    "description": "Generate structured mammogram report
        from features.",
    "parameters": {
        "type": "object",
        "properties": {
            "mass_shape": {"type": "string"},
            "microcalcification_pattern": {"type":
                "string"},
            "architectural_distortion": {"type": "string"},
            "breast_density_category": {"type": "string"},
            "malignant": {"type": "boolean"}
        }
    }
}
```

Function calling constrains the LLM to produce feature faithful structured output.



Mammogram Report

Run ID: 20260208_091621
Model: google/gemma-3-27b-it:free
Date: 2026-02-08T09:16:21.286291

MAMMOGRAPHY REPORT

Examination: Screening mammography.

Breast Composition: The breasts demonstrate heterogeneously dense tissue, which may limit detection of small masses.

Findings:

- **Mass:** A lobulated mass is present. Further characterization regarding size and location is not possible based on the provided information.
- **Microcalcifications:** No suspicious microcalcifications are identified.
- **Architectural Distortion:** No architectural distortion is present.

Impression: The identified lobulated mass warrants further evaluation, however, the absence of suspicious microcalcifications or architectural distortion is reassuring.

BI-RADS Assessment: BI-RADS 0 – Need Additional Imaging Evaluation.

Recommendation: Short-term follow-up with diagnostic mammography and/or ultrasound is recommended to further characterize the lobulated mass.

Notes

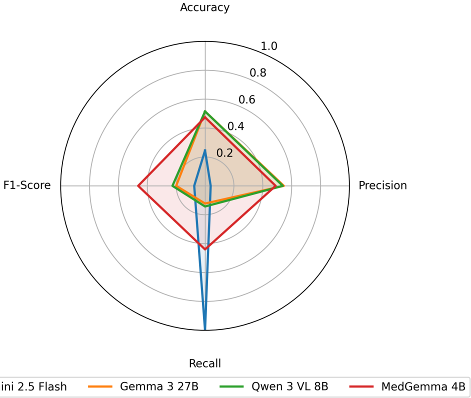
- For research use only

Example of an LLM generated mammogram report.

Model	Accuracy	Precision	Recall	F1-Score
Gemini 2.5 Flash	0.2449	0.0390	1.0000	0.0750
Gemma 3 27B	0.5152	0.5455	0.1224	0.2000
Qwen 3 VL 8B Instruct	0.5152	0.5385	0.1429	0.2258
MedGemma 4B	0.4742	0.4889	0.4400	0.4632

Performance comparison of LLMs on mammogram feature classification

Results & Discussion



Radar chart showing performance of different models across Accuracy, Precision, Recall, and F1-Score

Results & Discussion

- LLMs generated coherent reports reflecting structured features.
- Different models show different levels of accuracy under the same setup.
- Smaller LLMs outperform larger ones in most performance metrics for mammogram feature classification.
- Models tend to struggle with false positive cases, indicating difficulty in correctly identifying benign findings.
- This study highlights both strengths and limitations of current LLMs for structured clinical reporting.

Further Research & Limitations

Further research areas:

- Extend to multilingual report generation.
- Incorporate personalized analysis based on individual patient characteristics.
- Tracking changes over time by comparing mammogram features across multiple exams.

Current limitations:

- Dependency on feature annotation quality and current LLM hallucination risk.

References

- Mammograms: What You Need to Know. *Breastcancer.org*. `https://breastcancer.org/screening-testing/mammograms`
- Specific microcalcification patterns are associated with higher risk of breast malignancy. *Diagnostics* (2025).
- BI-RADS features including density and morphology guide mammographic reporting. *ScienceDirect Topics* (2022).
- An entity extraction pipeline for medical text records using large language models. *J Med Internet Res* (2024).
- MedGemma framework and evaluation for medical images and text. *arXiv Preprint* (2025).

Thank you for your attention

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